Loan Dataset

Data analysis and verification for machine learning

Introduction

Background

- Dataset from Kaggle.com
- Fully synthetic
- Cleaned in SQL
  - Removed negative values
  - Removed rows with nulls that had no application

Purpose

- Verify dataset can be used for machine learning
- Establish approval factor correlations

Data

Demographical

- City
- Age
- Education
- Gender
- Employment type

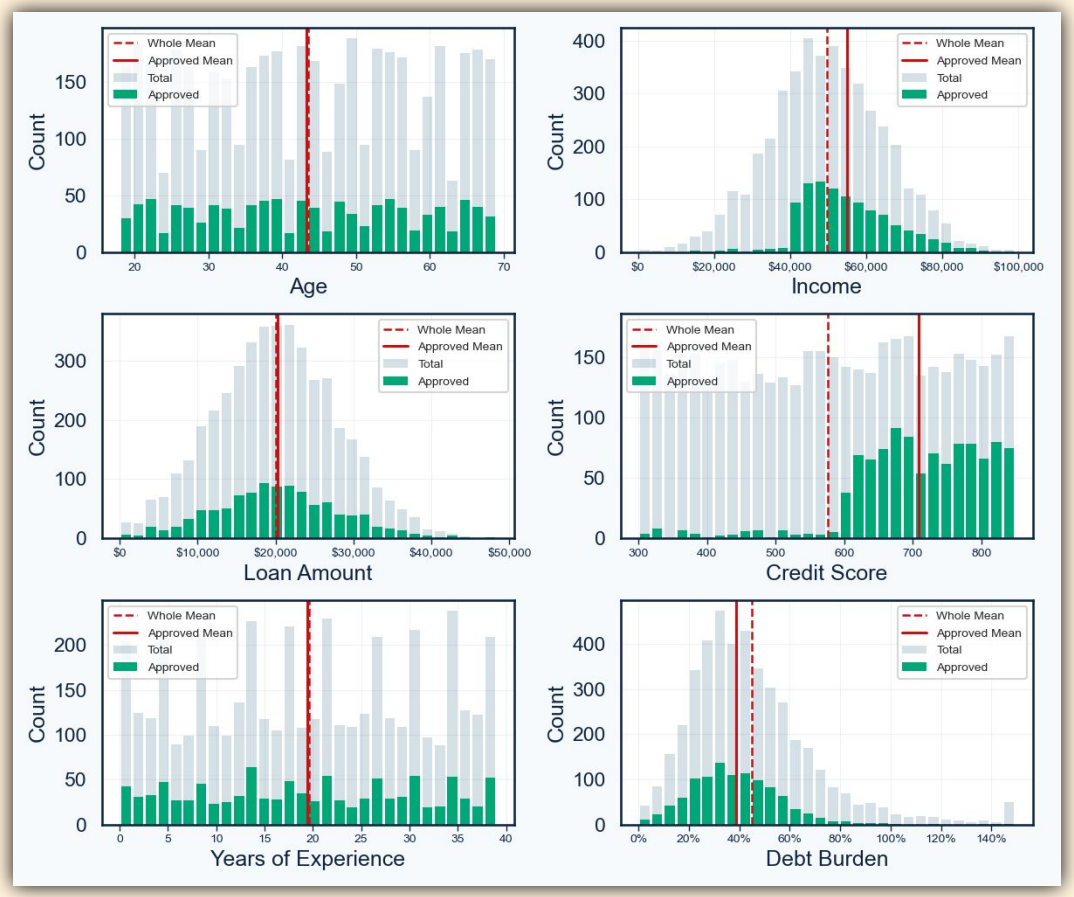
Numerical

Raw data:

- Loan Approval
- Loan amount
- Income
- Credit Score
- Years of work experience

Calculated:

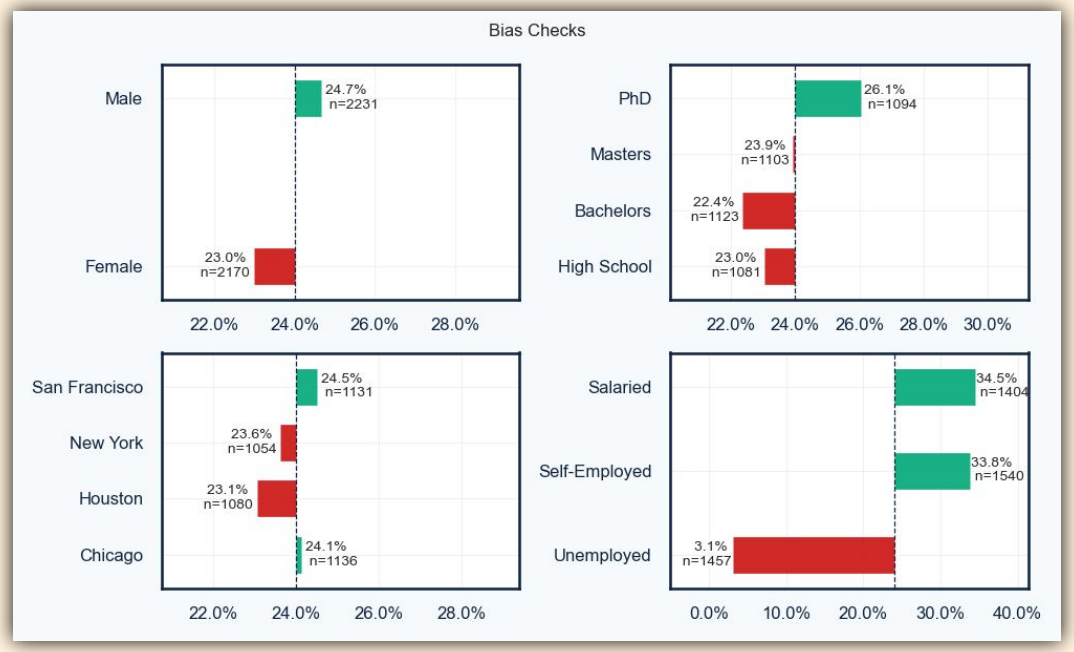
- Debt burden



Demographics & Variables

Each bin:

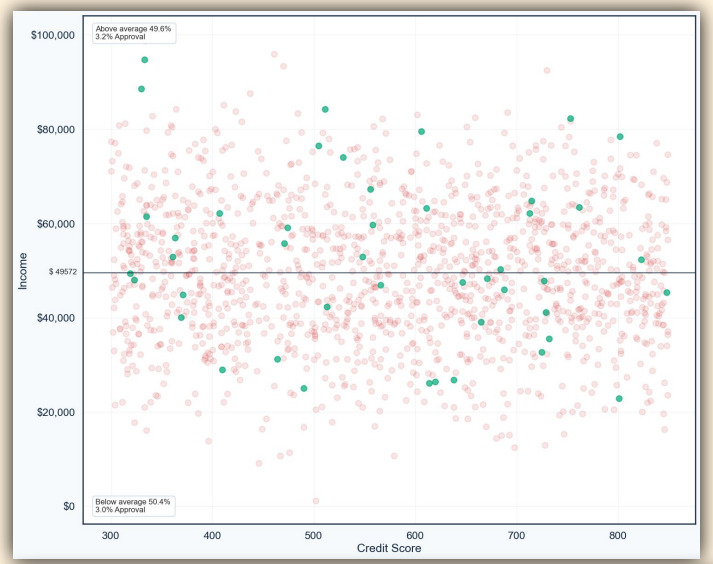
- Total number of applicants
- Total number of approved applicants

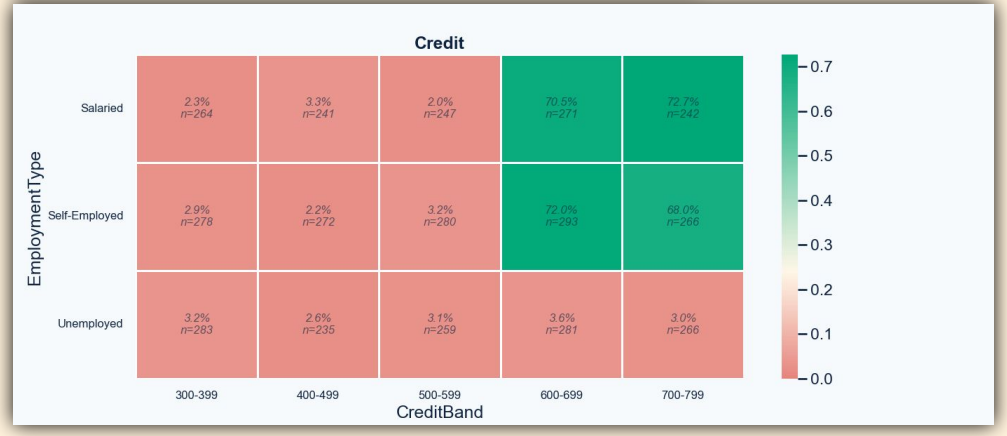


Bias Check

Portfolio target approval is 24%. Any data within 1% of the target is considered non-bias.

We notice a trend in bachelor’s level education and the employment types.



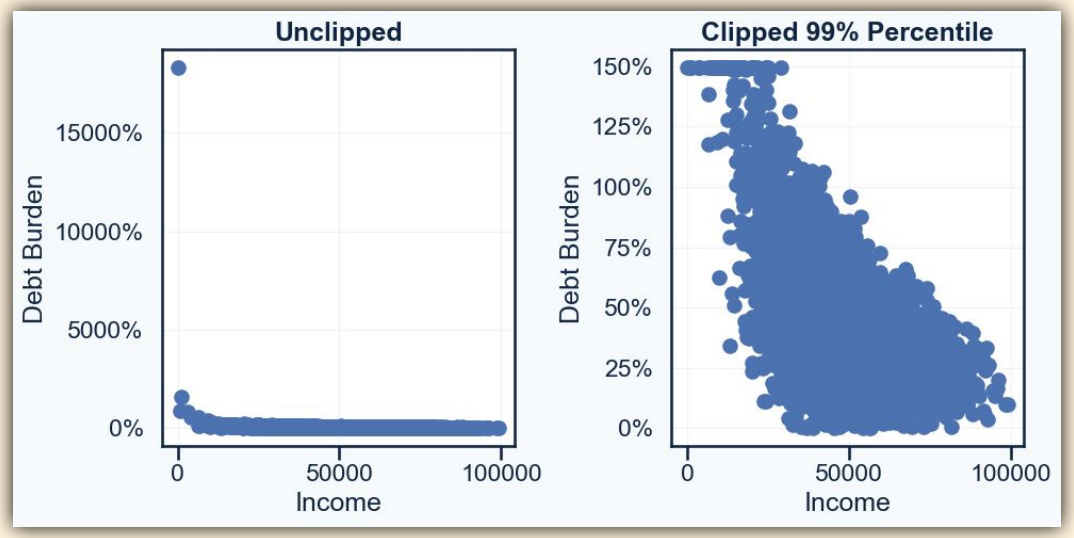


Unemployed Check

Top plot green points indicate loan was approved & applicant was unemployed

Bottom plot correlation heatmap between credit score bands and employment type

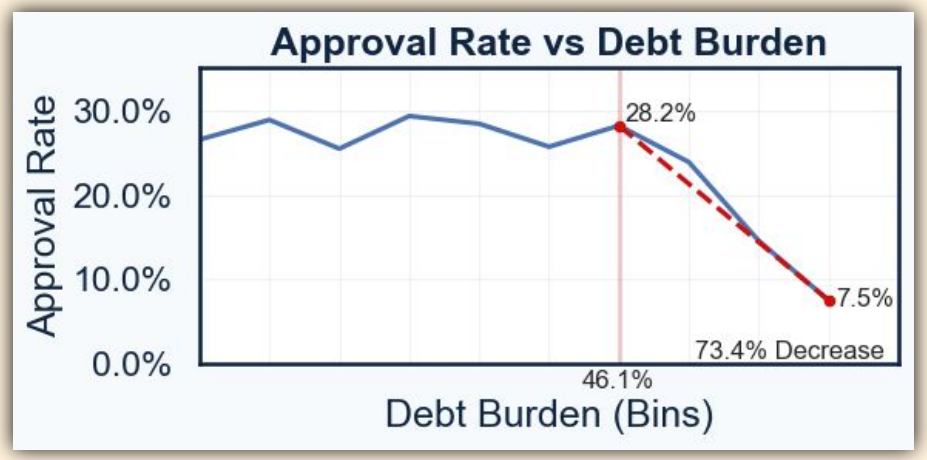
No noticeable correlation with unemployment from the data. Likely indicates unemployment is a direct factor.



Debt Burden

As the dataset does not give us a total debt amount for an applicant, we will take total debt as the loan amount

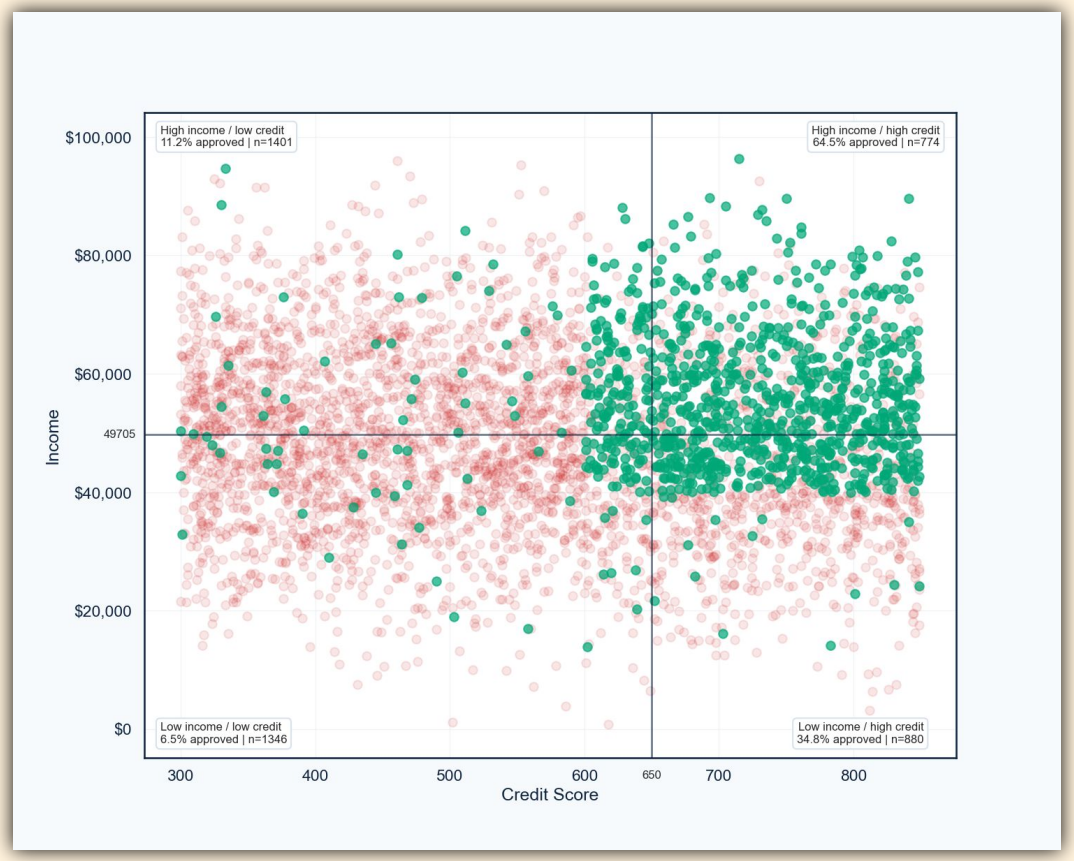
We clip the data for exploratory reasons and bring down severe outliers



Debt Burden Quantiles

Tail has a hefty approval rate decline in the most leveraged debt burden segments

Likely a secondary filter for loan approval



Credit Score v Income

Quadrants setup by 650 credit score and mean income of the dataset

Noticable trend below each line at approx 600 credit score and $40,000 income

Significant approval rate jump above 600 credit score and above $40,000 income

Correlation

Using Pearson Correlation, we verify that credit score is the biggest approval factor followed by income

Dataset appears to be fit for machine learning

Assesment

- Credit score is the highest impact factor
  - Significant jump in approval above 600 credit score
- Employment seems to be a clear secondary filter
  - Unemployed applicants are very rarely approved regardless of income and credit score
- Income is a slight factor, but coincides with credit score
- Debt burden acts as a secondary filter
  - Higher quantiles of debt burden likely to **NOT** be approved

Future Work

Work with additional data to:

- Find correlations between data and approval
- Risk assessment rather than binary approval
- Loosen or tighten approval based on risk

Research and understand financial and statistical calculations & topics that could be applicable